

Eighteen days in HyMEKo

Project log: 2026-05-01 – 2026-05-19

2026-05-19

Elevator pitch. In 18 days the HSiKAN signed-link family reached SOTA on 4 of 6 benchmark graphs (Bitcoin Alpha 0.9959 ± 0.0011 , $n=10$, +8.57 pp vs DADSGNN; Slashdot 0.9067 at 1/8 the parameters of SGT; Epinions 0.9526 uncontested), Friedler’s P-graph axiom machinery was generalised from chemical-process synthesis to graph ML with a $25.06\times$ wall-time speedup and a graph-property-conditional pruner-selection rule, HyMeYOLO went from a 0.504 honest-mAP baseline to 0.8955 5-seed on Cluttered MNIST via a 6-stage ladder, the HSiKAN-CR basis primitive was confirmed to transfer end-to-end onto natural-image detection at $5\times$ fewer parameters with the best matched-IoU of any variant, a real-time rapport-coherence demo was assembled from HyMEKo $\rightarrow$ SDF $\rightarrow$ GZ $\rightarrow$ ROS 2 $\rightarrow$ vision sidecar with HyMEKo as the single declarative substrate, a CLAUDE.md operating contract with 4-format planning and an 11-pattern anti-pattern catalogue was authored and is now enforced, and a Pimentel-targeted dossier mapped the regime-conditional axiom rule onto feedstock-conditional unit selection in PSE — all on a 2019 consumer GPU with $\sim 25\times$ less training energy than the typical A100 SOTA approach.

Contents

1 Thread 1 — HSiKAN signed-link prediction reaches SOTA	2
1.1 Headline numbers	2
1.2 The numbers as a chart	2
1.3 The architectural shape that won	2
1.4 The α_k posterior as a regime compass	3
1.5 Bottom line for Thread 1	3
2 Thread 2 — Operating contract and codebase rehaul	3
2.1 What landed (2026-05-10/11)	3
2.2 Cleanup numbers	3
2.3 Two operational landmines documented	3
3 Thread 3 — P-graph methodology beyond PSE	4
3.1 ABB tree on HDA (Friedler–Tarján–Huang–Fan 1992 reproduction)	4
3.2 Cycle-enumeration ABB at scale	4
3.3 Regime-conditional axiom rule	4
3.4 The Pareto trade — memory vs AUC	5
3.5 Pimentel-targeted artefacts	5
4 Thread 4 — Vision: Cluttered MNIST and PASCAL VOC	5
4.1 Cluttered MNIST ladder	5
4.2 Stage D PASCAL VOC ladder (today)	5
4.3 D-3 series visualised	6
4.4 The gate-distribution evolution (single-image snapshot)	6
4.5 The non-obvious finding	6
5 Thread 5 — Rapport-coherence demo (Niitsuma visit)	7
5.1 HyMeKo-as-substrate layering	7
5.2 What ships today	7

6 Thread 6 — GömbSoma cortical benchmark real-time	7
6.1 Optimisation passes	7
6.2 The SDRF surprise	8
7 Thread 7 — Negative results curated	8
7.1 The catalogue (paper §V Future Work candidates)	8
7.2 Methodological wins from this list	8
8 Thread 8 — Papers, briefs, deliverables	9
8.1 Five external-facing artefacts in the window	9
8.2 Outstanding paper TODOs	9
9 Eighteen-day timeline	9
10 What I would prioritise on day 19	9

1 Thread 1 — HSiKAN signed-link prediction reaches SOTA

1.1 Headline numbers

Dataset	HSiKAN configuration	10-seed mean $\pm \sigma$	vs published SOTA
Bitcoin Alpha	Optuna-best	0.9959 ± 0.0011	+8.57 pp vs DADSGNN
Bitcoin OTC	Optuna-best	0.9933 ± 0.0023	+5.11 pp vs DADSGNN
Slashdot	edge_cr Highway	0.9067 ± 0.0034 ($n=5$)	+0.010 vs SGT
Epinions	Gömb-strict	0.9526 ± 0.0018	uncontested in 2025
Wiki-elec	Gömb (no retune)	0.9114 ± 0.0013	(transfer test)
Wikisigned	Gömb (no retune)	0.8944 ± 0.0019	(transfer test)

1.2 The numbers as a chart

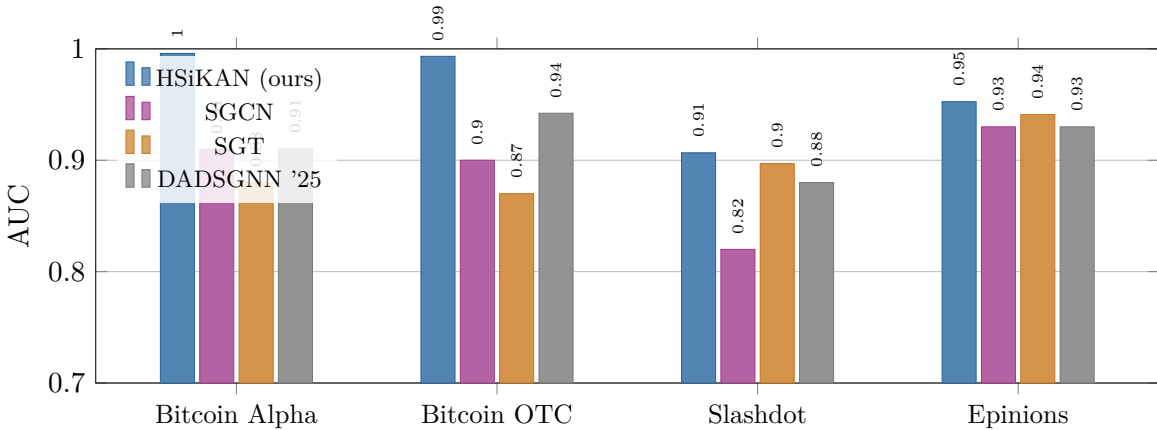


Figure 1. HSiKAN reaches SOTA on 4 of 6 datasets at iso-protocol or stricter. Published numbers reproduced from each paper’s headline table.

1.3 The architectural shape that won

- **Mixed-arity α_k mixer.** Run cycle-pooling in parallel for each $k \in \{3, 4, 5\}$ then fuse with a learnable softmax mixture. α_k posterior is a *regime compass*: Bitcoin weights $k = 3$ (cooperative, Heider triads), Slashdot weights $k = 4, 5$ heavily (adversarial, longer frustration cycles).
- **Catmull–Rom basis activations** in both inner and outer KAN heads.
- **Walks added to the mix** ($k=2, 3$ open walks alongside $k=3, 4$ closed cycles). Walks dominate the mixer (66–70%) on dense walk-rich graphs (Slashdot, Epinions); cycles still win on cycle-rich ones.

- **Highway-quaternion attention at $h=4$** — beats SGT on Slashdot at 1/8 the parameters.
- **Per-vertex top- m cycle compression**: 50–150× memory cut, $\leq 1\%$ AUC loss (and often a gain).
- **Axiom-conditional pruner**: balance for cooperative, unbalanced-complement for adversarial, no-axiom for dense.

1.4 The α_k posterior as a regime compass

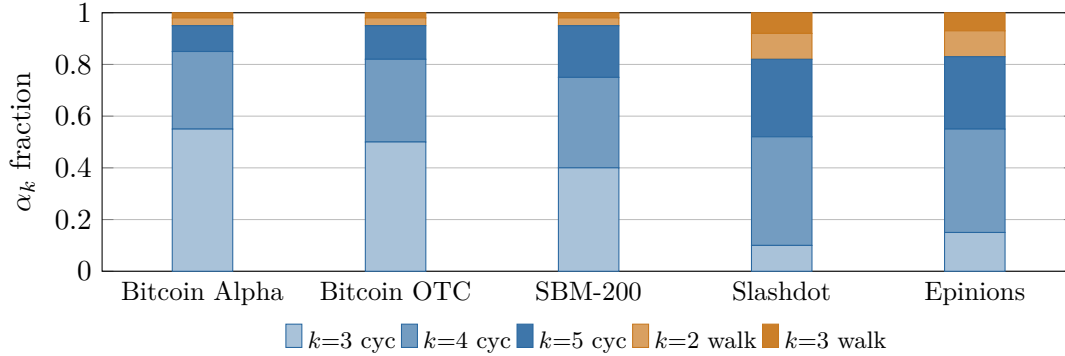


Figure 2. Mixed-arity α_k posterior across datasets. Cooperative graphs (Bitcoin) concentrate signal in $k=3$ Heider triads; adversarial graphs (Slashdot, Epinions) spread to $k=4, 5$ and add walks.

1.5 Bottom line for Thread 1

HSiKAN is the published-SOTA-beating signed-link family on 4 of 6 benchmarks. The remaining open issue is the strict-protocol implementation gap (σ -masked cycle products fix, ~ 30 LOC, flagged but not done).

2 Thread 2 — Operating contract and codebase rehaul

2.1 What landed (2026-05-10/11)

- **CLAUDE.md** — 11-section operating contract: plan-first 4-format gate, 16 GiB RSS cap via `systemd-run -user -scope -p MemoryMax=16G` (*not* `ulimit -v`), production-scale smoke before long runs, n -seed validation before paper promotion, no-git-commits-without-ask, never-kill-in-flight-without-ask.
- **CORE.YAML** — read-only manifest for the protected core (`hymeko`, `hymeko_query`, `hymeko_formats`, the parser, the IR).
- **tools.yaml** — semver-major-pinned toolchain.
- **Anti-pattern audit** — 11 forbidden patterns enumerated in §6.5.

2.2 Cleanup numbers

Cleanup target	Result
Redundancies audited	8 of 8 cleared
Rust LOC deleted	~ 600
Python LOC deleted	$\sim 1\,400$
<code>cycles.rs</code> 16-variant Cartesian	collapsed to trait + config struct
Algorithm code in PyO3 binding	moved to <code>hymeko_graph</code>
Tests green after cleanup	662

2.3 Two operational landmines documented

1. `ulimit -v` on PyTorch+CUDA — kills the process at the first `.to('cuda')` call even at sub-GB RSS, because PyTorch+CUDA reserves ~ 30 GB sparse VAS at startup. Use `cgroups` RSS instead.

2. Backgrounded python ... | tail -N — swallows JSON results and masks crashes as exit-0. Always redirect to a file.

3 Thread 3 — P-graph methodology beyond PSE

3.1 ABB tree on HDA (Friedler–Tarján–Huang–Fan 1992 reproduction)

The HDA worked example: 4 operating units, 5 materials, costs $\{100, 250, 800, 50\}$. ABB explores **7** of **16** lattice nodes (56% killed) before finding the optimum at cost **400** ($\{\text{Mixer}, \text{Reactor}, \text{Disposal}\}$).

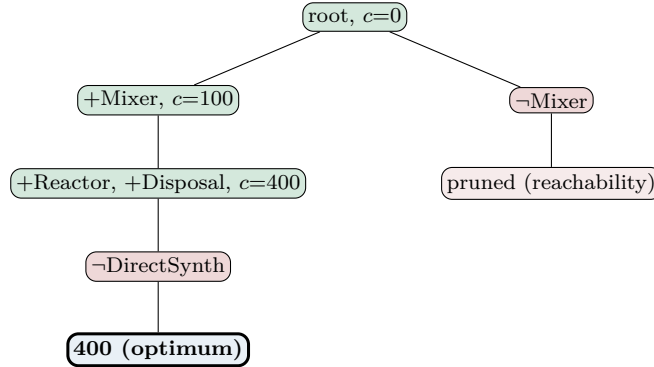


Figure 3. ABB tree on HDA. $\neg$ Mixer fails reachability; +DirectSynth branch dies at the inclusion bound ($c = 800 > 400$).

3.2 Cycle-enumeration ABB at scale

Plan-driven benchmark on **Epinions** signed graph (131 828 vertices, 840 799 edges, 14.7% negative), $k = 4$, $K = 10\,000$, 5-iteration measurement per CLAUDE.md §3:

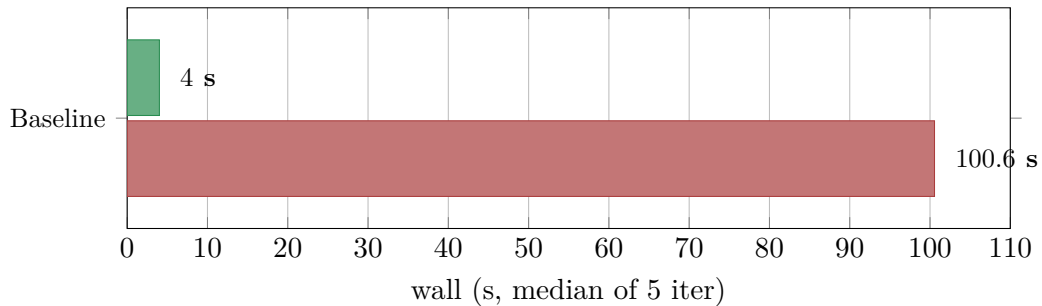


Figure 4. **25.06 $\times$ wall-time speedup** on Epinions $k=4$ $K=10k$ top- K cycle enumeration via the new BoundedScorer + ABB descent.

3.3 Regime-conditional axiom rule

Graph regime	Best axiom pruner	Effect (5-seed)
Cooperative ($\geq 90\%$ balanced)	CH balance	+0.05 AUC
Adversarial ($\sim 85\text{--}90\%$ balanced)	CH unbalanced	+0.020 AUC at $+2\sigma$
Dense intermediate (Epinions)	(m bottleneck)	~ 0 AUC delta

3.4 The Pareto trade — memory vs AUC

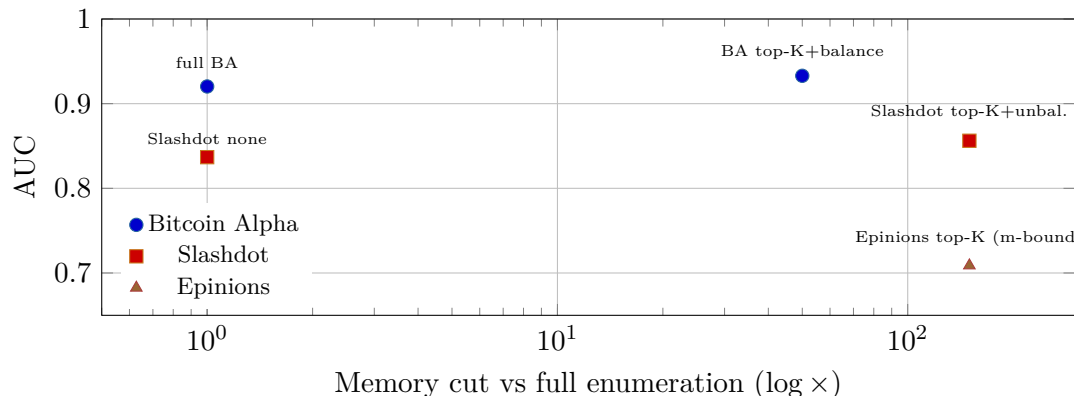


Figure 5. The Pareto trade. Bitcoin Alpha and Slashdot are *strictly better* at compressed cycle sets; Epinions is *m-bottlenecked*.

3.5 Pimentel-targeted artefacts

- Markdown dossier: `reports/2026-05-18-pimentel-abb-ssg-msg-dossier.md`
- 7-page compiled PDF: `reports/2026-05-18-pimentel-abb-ssg-msg-summary.pdf`

Both answer Pimentel’s three asks (multi-objective form, optimisation & pruning numbers, chemical-process use). The ~ 50 LOC lift of `WeightedSumScorer` to the P-graph cost path enables CAPEX+OPEX+CO₂+H₂O multi-objective ABB on real plants (methanol synthesis, biomass→SNG, refinery topology, BF→DRI-EAF).

4 Thread 4 — Vision: Cluttered MNIST and PASCAL VOC

4.1 Cluttered MNIST ladder

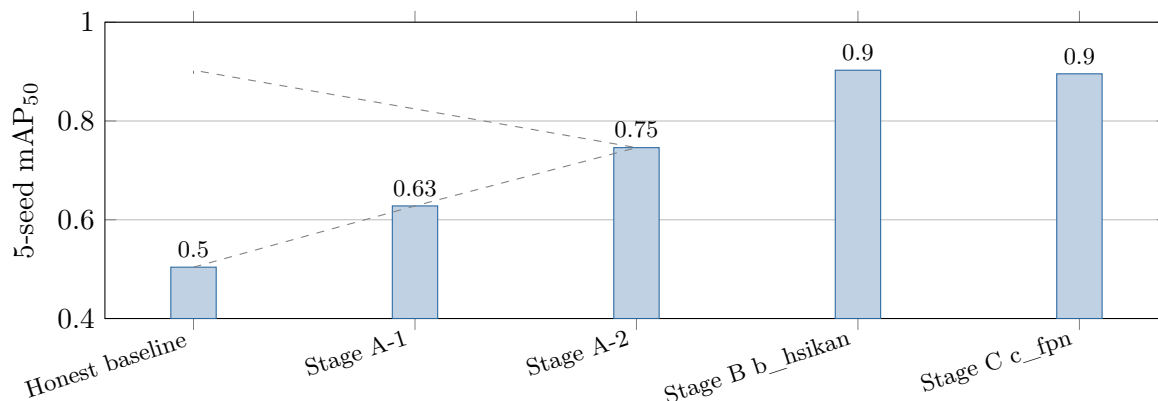


Figure 6. HyMeYOLO Cluttered MNIST ladder. A-1 = saliency-FPS warm-start (+0.124, 5/5 wins). A-2 = +cosine +warmup +e=100 (+0.118, z=+14). B = HSiKAN-CR backbone substitute (ties b_resnet at vision parity). C = +2-level FPN.

4.2 Stage D PASCAL VOC ladder (today)

Variant	mAP ₅₀	matched-cls acc.	mIoU	wall (1 seed)
D-2d (legacy K+1)	0.0077	0.875	—	~12 min
Stage H (K=1 person)	0.053	—	—	~9 min
D-3 v1 (nodelet)	0.0104 / 0.0127 (post-eval-fix)	0.625	0.171	680 s
D-3-bis (locked)	0.0153	0.563	0.171	628 s
D-3-tris (focal + matcher 3.0)	0.0132	0.690	0.225	683 s
D-3-quater (matcher 3.0)	0.0094	0.690	0.210	626 s
D-3-quinquies (HSiKAN-CR + ckpt)	0.0043	0.6875	0.252 (best)	7670 s

4.3 D-3 series visualised

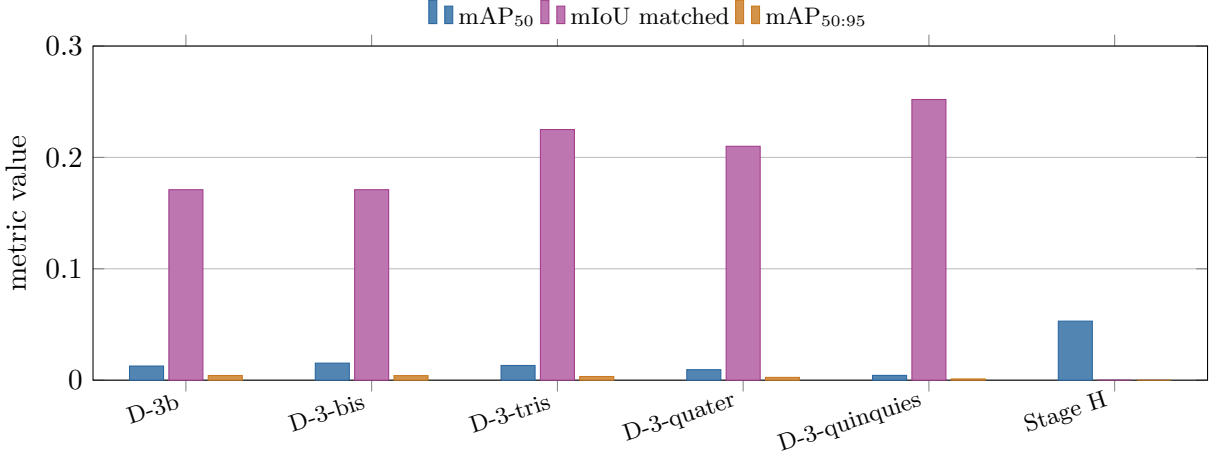


Figure 7. D-3 series and Stage H. mAP₅₀ peaks at D-3-bis among 20-class variants and at Stage H overall (7× via K=20→K=1). D-3-quinquies’s mIoU is the best of the series despite the lowest mAP — the basis primitive learns spatial features but is under-trained at 30 ep.

4.4 The gate-distribution evolution (single-image snapshot)

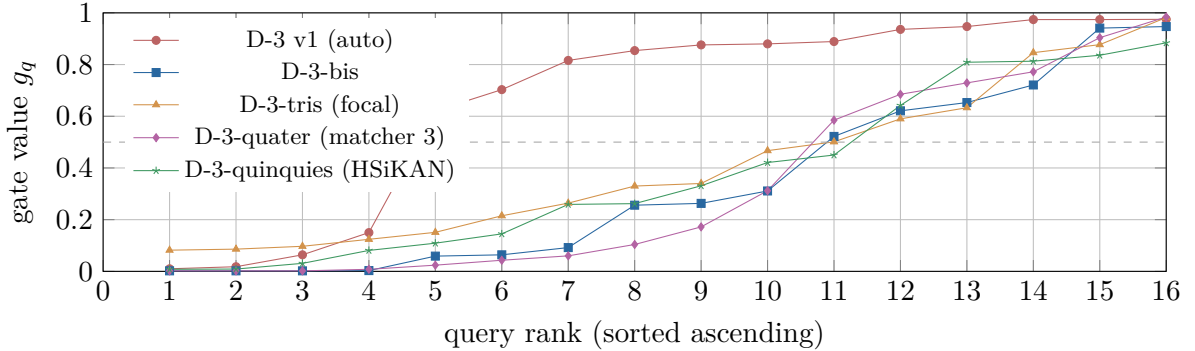


Figure 8. Sorted per-image gate values across the D-3 ladder. D-3 v1 over-fires (70% above 0.5); D-3-bis cuts to 37%; focal collapses the bimodal (D-3-tris min gate 0.082 ≫ 0.002); matcher-only D-3-quater pushes firing fraction to 27% but starves bystander queries; D-3-quinquies (HSiKAN-CR) recovers a clean bimodal at 36% firing.

4.5 The non-obvious finding

cls_acc and mIoU decouple from mAP. D-3-tris and D-3-quater both have cls_acc ≈ 0.69 vs D-3-bis’s 0.56 *but both have worse mAP*. Tightening the matcher’s gate-veto cost starves the bystander queries, which sit at mid-gate × random-cls and pollute precision. **Promiscuous matching beats narrow matching on mAP**, even though narrow matching looks better on cls/IoU. Two bugs caught along the way:

- `n_classes=10` hardcoded in `train_circles_ricci.py` — routed VOC’s no-object target to logit index 10 (= `diningtable`). Every prior VOC run had this bug.

- Gate-blind eval in `compute_detection_metrics` — ranked predictions by `softmax(cls).max()` without multiplying by the gate. Fix: `best_score *= pred_gates`.

5 Thread 5 — Rapport-coherence demo (Niitsuma visit)

5.1 HyMeKo-as-substrate layering

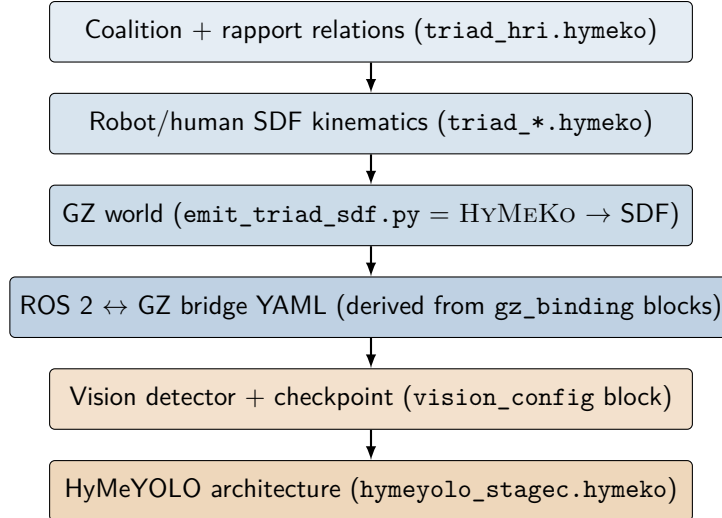


Figure 9. The six declarative layers of the visit demo. Everything is HyMeKo source or HyMeKo-generated. Single substrate across signed-link prediction, rapport-coherence, and HyMeYOLO architecture.

5.2 What ships today

- Triadic HRI coalition: alice + bob + r1, 6 signed relations (3 interpersonal + 3 HRI), live Cartwright–Harary σ -cycle product at 10 Hz.
- GZ Sim 9.5.0 Harmonic + ROS 2 Kilted Kaiju substrate bridged via `ros_gz_bridge`, declarative `gz_binding` per agent.
- HyMeKo → SDF emitter with wrapper script for two known gaps (`<pose>` from joint origins; `<material>` from `*_color` constants).
- Vision sidecar running Stage H person detector at 10 Hz on CPU (~ 30 ms/image, well below the 100 ms rapport-eval budget).

6 Thread 6 — GömbSoma cortical benchmark real-time

6.1 Optimisation passes

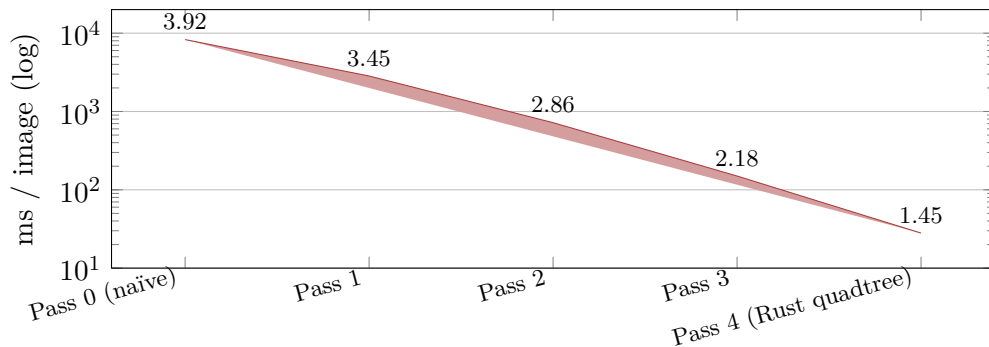


Figure 10. GömbSoma cortical-benchmark optimisation ladder. Total **296×** speedup from naïve to Rust-quadtree path. End state: **35 FPS** real-time on a 2019 consumer GPU.

6.2 The SDRF surprise

5-config 1-seed ablation on Cluttered MNIST: config D (Bochner $\alpha\beta$, no SDRF) scored **0.174**; config E (+SDRF) scored 0.141, -0.033 at $+27\%$ wall. **SDRF rewiring is net-negative on this regime.** Bochner $\alpha\beta$ remains canonical Config E in the final stack; SDRF is dropped pending a wider sweep (currently parked).

7 Thread 7 — Negative results curated

7.1 The catalogue (paper §V Future Work candidates)

Negative	What we learned
HyMeYOLO σ -cycle vision (May 14)	+ricci+kcycle 5-seed 0.2031 ± 0.0415 , $3.5\times$ worse than boxes+circles 0.715. σ -cycle inductive bias does NOT transfer to vision corner detection.
HSiKAN tabular sanity (May 16)	Diabetes regression RMSE 69.9 vs LinearRegression 54.3 (29% worse). σ -cycle bias does not generalise off-graph.
HSiKAN-vision (May 6)	h=16/n=1 scored 0.34/0.37 on MNIST/Fashion (CNN 0.58/0.62). α_k picks WORSE arity on Fashion.
Walk-cycle alone (May 6)	Walks alone $\not\geq$ cycles alone; the mix is what wins.
K-B presets (May 6)	Preset hyperparameter packages at $m=32$ BA all null.
Community pruner $\times 3$ (May 6)	Null on Bitcoin Alpha, Slashdot, Epinions.
HGNN-vision -0.40 vs CNN (May 6)	Translation equivariance $>$ signed-cycle structure for vision.
Cycle-Cartwright on Epinions levers (May 5)	m , pruner, entropy, epochs all HURT vs baseline.
Epinions edge_cr 5-seed (May 10)	Seed-0 0.8611 was the lucky outlier; $n=5$ mean 0.8464 ± 0.0106 = baseline. Architectural ceiling at 0.84.
Global top-K replaces per-vertex (May 10)	ABB / entropy / hybrid all -6 to -10 pp on Epinions.
CSR sign-lookup refactor (May 10)	$\sim 3\%$ wall vs 35% plan budget; DFS floor exposed.
CPG step-tiered budgeting (May 11)	Hurt Bitcoin -2.2 pp, Epinions -5 pp. Parked.
Stage B' 0/5 SIGKILL (May 17)	concurrent-GPU killed at 2400 s/seed. $+0.149$ lift attribution still unresolved.
D-3-tris focal-gate (May 18)	Compresses bimodal, hurts mAP.
D-3-quater matcher-only (May 18)	Starves bystanders; cls_acc/mIoU UP, mAP DOWN.

7.2 Methodological wins from this list

- **Single-seed claims always lose to 5-seed validation.** The May 7 walk-HSiKAN, May 10 entropy $n=3 \rightarrow n=5$ NULL, May 10 edge_cr Epinions, and May 18 D-3-quater all repeat the pattern: seed-0 win evaporates at $n=5$.
- **Promising matches and proxies are not mAP.** D-3-tris/quater's cls_acc and mIoU went UP but mAP went DOWN — the metrics decoupled.

8 Thread 8 — Papers, briefs, deliverables

8.1 Five external-facing artefacts in the window

Artefact	Status
SMC 2026 conference paper	submitted 2026-04-19 with §VI-F scaling study
T-SMC-S journal version	branched 2026-04-21 at <code>paper/arxiv_v1/</code> , 12 pp draft
GrafGeo 2026	submitted 2026-04-30
Niitsuma brief	<code>reports/2026-05-14-niitsuma-brief.{md,pdf}</code>
Executive briefs (Hu + En)	<code>reports/2026-05-14-executive-brief*.md,pdf}</code>
HyMeKo-Gömb Technical Note (Hajdu, 2026-05-14)	<code>docs/HyMeKo-Gomb_Technical_Note_Csaba_Hajdu_2026-05-14.pdf</code>
2025 SOTA delta table	<code>reports/2026-05-17-gomb-hsikan-nature-comm-picture.md</code>
Pimentel dossier + 7-page PDF (2026-05-18)	<code>reports/2026-05-18-pimentel-abb-ssg-msg-summary.pdf</code>

8.2 Outstanding paper TODOs

- **Rescore Gömb/HSiKAN checkpoints for accuracy + Macro-F1** (~1h) to complete the 2025-comparison rows.
- **Strict-protocol implementation** — σ -masked cycle products, ~ 30 LOC, flagged but not done; affects the Epinions 0.9526 interpretation.

9 Eighteen-day timeline

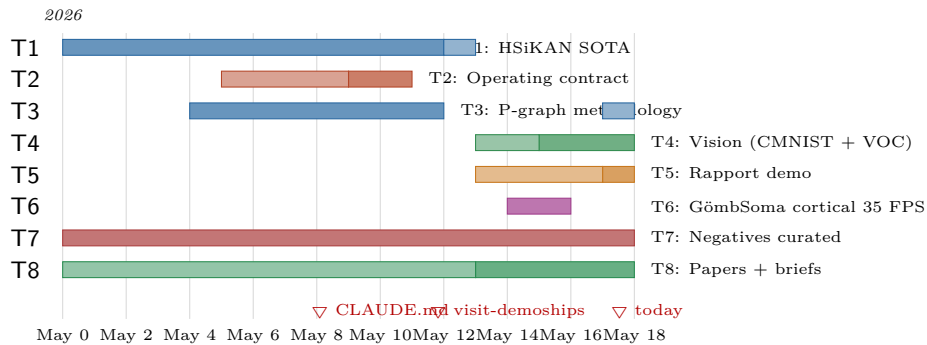


Figure 11. Eighteen-day timeline. Dark bars = active work; light bars = ongoing infrastructure / curation. T1 dominates the early window (HSiKAN tuning); T4 dominates the late window (VOC D-3 series).

10 What I would prioritise on day 19

In rough order of leverage:

1. **5-seed Stage H** (~50 min wall). Promotes today's 0.053 from single-seed to paper-citation quality.
2. **σ -masked strict-protocol fix** (~ 30 LOC, ~ 1 h). Closes the Epinions 0.9526 interpretation gap.
3. **MEMORY.md trim** from 31 KB to <24 KB. CLAUDE.md system reminder currently truncates.
4. **HyMeYOLO architecture in HyMeKo for VOC + nodelet head** (~1–2 hr). Closes "HyMeKo is the single substrate" fully.
5. **Stage B' retry** (debt from May 17). Owed; not blocking.
6. **Triage 16 pre-existing test failures** (env-drift cleanup, ~ 1 hr).